DPP	DEPARTMENTAL POLICY AND PROCEDURE		
Title	**CAPILLARY BLOOD GAS (CBG) SAMPLING**		
Department	**Neonatology Department**		
Applies to	**All NICU Medical and Nursing Staff**		
Policy Code and Version Number	**DPP-NICU-005 (v2)**		
Last Version Approval Date	**18/2/2021**	Number of Pages	**4**

	NAME	POSITION	SIGNATURE	DATE
Prepared by:	**MS. JOICY JOY**	NICU Quality Representative		27/2/2024
	MS. AISHA ZAIBI SAAD AL-SHUMRANI	Neonatal Intensive Care Unit, Head Nurse		27/2/2024
	DR. TAYSEER HASSAN	Neonatology Resident		27/02/2024
Reviewed by:	**DRA. FAWZIA MOHAMMED FARAHAT**	Head, Neonatology and Pediatric Department		14/03/24
	MR. MANSOUR AHMED AL-ZAHRANI	Head, Quality Management and Patient Safety		14.3.24
Approved by:	**MR. SULTAN ALI AL-GAMDI**	Nursing Director		19-3-24
	DR. WAEL KAMEL ARBA	HED Assistant for Medical Affairs		19.3.24

APPROVAL DATE	EFFECTIVE DATE	TYPE OF REVISION	REVIEW DATE
19 / 3 / 2024	7/ 4 / 2024	TRIENNIAL	18 / 3 / 2027

Baha_Cluster
Healthcluster-baha@moh.gov.sa
مستشفى المخواة العام
Al-Mikhwah General Hospital

1.0 PURPOSE

1.1. To obtain a capillary blood gas that will provide accurate laboratory results with minimal discomfort and potential for injury/infection.

1.2. To check blood gas for patient level of oxygenation.

2.0 DEFINITION/ABBREVIATION

2.1. **NICU** - Neonatal Intensive Care Unit

2.2. **Capillary Blood Gases (CBG)** will be obtained to assess adequacy of oxygenation and ventilation to infants who do not have arterial line access, per physician's order.

3.0 RESPONSIBILITY

3.1. **NICU Physician** (Residents, Specialist) are responsible in providing a written order in the physician's order sheet.

3.2. **NICU Nurses** are responsible in obtaining blood gas through capillary route.

3.2.1. The nurse should relay CBG report to physician.

4.0 POLICY

4.1. Blood gas is a laboratory test of blood where the purpose is primarily to measure ventilation and oxygenation. The source is either from arterial blood gases, venous blood gases and capillary blood gases. Use a pre-heparinized capillary tube to collect the blood from the heel of the newborn, blood should flow freely from the puncture site.

4.2. Capillary blood gas (CBG) must be done by a nurse with a physician's written order.

4.3. Nurses must be aware of when and how to relay critical CBG report to physician.

4.4. All CBG report must be documented in the patients' medical file.

4.5. Standard precaution must be observed. Blood contaminated items should be disposed according to infection control policy.

4.6. Blood gas machine must be checked that is functioning before extracting sample for CBG.

4.7. The heel of the newborn must be warmed 10-15 minutes, to dilate the vessels in the area before the blood sample is taken to improve accuracy. Use socks bootees and extra blankets to facilitate warming. Direct heat is not recommended.

4.8. To prevent necrotizing osteochondritis of the underlying oscalcis (heel bone) the puncture should be made on the fleshy outer aspect of the heel (hatched area).

4.9. If alcohol is used to cleanse the outer aspect of the heel of an infant, it must be dried roughly before the preferred site is lanced and the first drop of blood must be discarded.

4.10. Pressure should be applied briefly to the puncture site after collecting blood sample to prevent painful extravasations of blood into subcutaneous tissues.

4.11. When inserting the sample in the blood gas analyzer put the side of the capillary tube which has not been in contact with the infant's foot as this can affect the blood gas analyzer.

4.12. Capillary blood gas sampling should NOT be performed at or through the following sites:

4.12.1. Presence of posterior curvature of the heel as the device may puncture the bone.

4.12.2. The fingers of neonate to avoid nerve damage.

4.12.3. On previous puncture sites

4.12.4. Cyanotic or poorly perfused tissues

4.12.5. Localized areas of infection

5.0 PROCEDURE (S)

5.1. Perform hand hygiene to prevent nosocomial infection.

5.2. Identify the infant by checking the medical record and Identification band.

5.3. Assemble equipment and check the functional of Blood gas machine to promote efficiency of the procedure.

5.4. Explain procedure to the parents if present to promote cooperation, and sense of involvement and control.

5.5. Warm the newborn's heel, 10 to 15 minutes prior to procedure.

5.6. Provide comfort measures such as non-nutritive sucking, facilitated tucking or swaddling

5.7. Don gloves.

5.8. Cleanse site (lateral and medial posterior surface is used) with alcohol swab.

5.9. Position the hand with fingers along the calf and thumb at the ball of the foot to stabilize. Apply pressure along the calf toward the heel.

5.10. Lanced the site and wipe away the first drop of blood.

5.11. Collect blood in capillary tube making sure that no air bubbles present, fill the blood gas tube to a **minimum of ¾ full**, holding the tube horizontally. Obtain a continuous flow to avoid clotting, cap the ends.

5.12. Place in ice to send to the laboratory or do the Blood gas analysis in blood gas machine in the Unit as soon as possible.

5.13. Apply pressure to the site with gauze and continue comfort measures.

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5.14. Notify the physician of the capillary blood gas results.

5.15. Document the procedure and the keep the capillary blood gas report in the patient's medical file.
5.15.1. Date and time test drawn, site of obtained specimen and infant's response to Procedure

6.0 EQUIPMENTS

6.1. Gloves.
6.2. Heparinized capillary tubes with caps
6.3. Alcohol swabs and gauze
6.4. Sterile lancet

7.0 ATTACHMENT/FORMS

7.1. Nurses Progress Notes (GDOH-NUR-NPN-203)
7.2. Physician Order Form (GDOH-INP-POF-079)

8.0 REFERENCES

8.1. Mhairi G. MacDonald, Jayashree Ramasethu, (2007) Atlas of Procedure in Neonatology (3rd Ed.). Lippincott Williams & Wilkins.

8.2. John P.Coherty, Eric C. Eichenwald, Ann R. Stark. (2008). Manual of Neonatal Care. (6th Ed.). Wolters Kluwer/ Lippincott Williams & Wilkins

8.3. Tricia Lacy Gomella, (2009). Neonatology Management Procedures, On-Call Problems, Diseases & Drugs. (4th Ed.). Appleton & Lange.

9.0 REVISION CLAUSE

9.1. Modification of policy format, lay out and policy coding

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KINGDOM OF SAUDI ARABIA

وزارة الصحة
Ministry of Health

Hospital: _____ مستشفى:

Region: _____ المنطقة/المحافظة:

Dept./Unit: _____ القسم/الوحدة:

MRN: | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

NURSES PROGRESS NOTES FORM

DATE &TIME	NURSES NOTES (D-data A-action R-response)	REGISTERED NURSE INITIAL WITH ID NUMBER

Note: Write the time in each entry & affix your initial at the end of each paragraph. Document your complete Name, Initial, Job number, Date & Time at the closure of your documentation. Draw a line across empty spaces.

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Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

PHYSICIAN ORDER FORM

ALLERGIC TO:

DATE	TIME	PHYSICIAN ORDERS	PHYSICIAN STAMP / SIGNATURE	RN INITIAL / JOB NUMBER